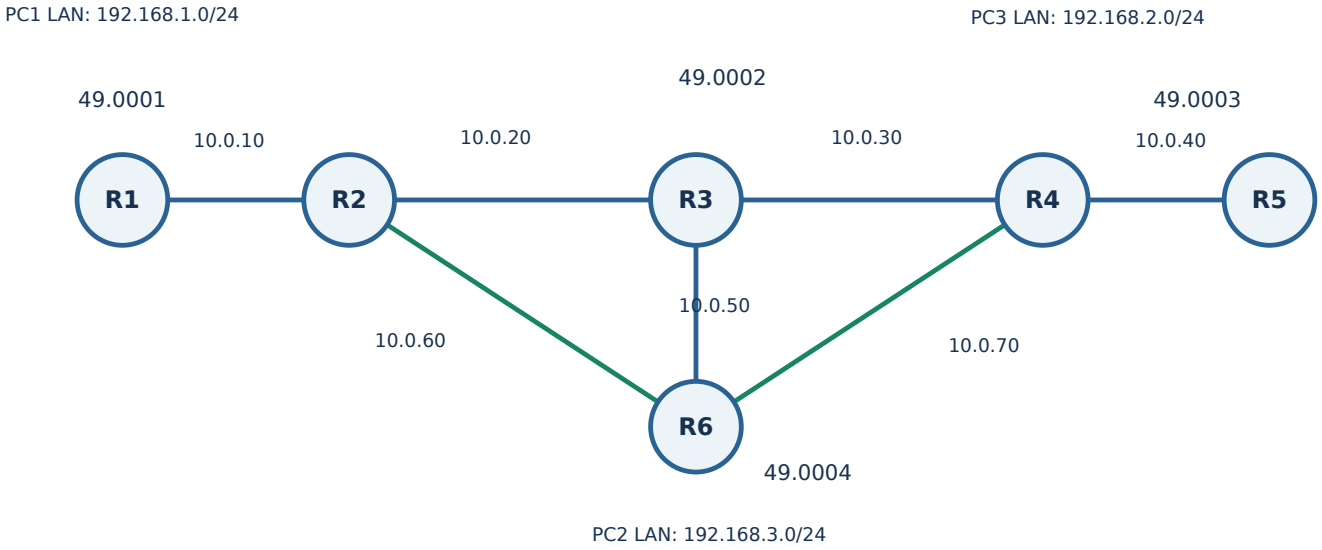
IS-IS Theory and Architecture

Six-router, four-area GNS3 laboratory | Level 1, Level 2, LSPs, SPF and redundant routing

Purpose. Explain how IS-IS works and connect every major concept to the actual R1-R6 GNS3 lab. This is a theory companion to the separate full-router configuration PDF.

Topology: main chain R1-R2-R3-R4-R5; redundant R2-R6-R4 and R3-R6 links



1. Router and area design

Router	Area / NET	IS-IS role
R1	49.0001 / ...0001.00	Level 1; LAN 192.168.1.0/24
R2	49.0001 / ...0002.00	Level 1-2; L1 toward R1; L2 toward R3/R6
R3	49.0002 / ...0003.00	Level 2 only; transit backbone
R4	49.0003 / ...0004.00	Level 1-2; L2 toward R3/R6; L1 toward R5
R5	49.0003 / ...0005.00	Level 1; LAN 192.168.2.0/24
R6	49.0004 / ...0006.00	Level 1-2; L2 toward R2/R3/R4; LAN 192.168.3.0/24

Note: The area is encoded in the NET, not in the local IS-IS process tag. Process tags such as KHALID-CORE and HARCHAOUI-CORE are locally significant.

2. Fundamentals: what is IS-IS?

Intermediate System to Intermediate System (IS-IS) is a link-state interior gateway routing protocol (IGP). Routers discover IS-IS neighbors, describe their links in LSPs, synchronize topology information, and run SPF to determine shortest paths.

The key objects

Object	Meaning in this lab
Intermediate System (IS)	An IS-IS router: R1, R2, R3, R4, R5 or R6.
Area	Groups Level 1 routing information. R1 and R2 share Area 49.0001; R4 and R5 share 49.0003.
NET	Network Entity Title. Example on R1: 49.0001.0000.0000.0001.00.
System ID	Unique 6-byte router identifier encoded in the NET, such as 0000.0000.0003 for R3.
NSAP selector / NSEL	Final NET byte, 00 for a router NET.
L1 / L2	Separate IS-IS flooding and SPF scopes, not OSPF area-number equivalents.

NET structure

49.0001 . 0000.0000.0001 . 00

49.0001 = area portion (area 49.0001); 0000.0000.0001 = unique system ID; 00 = NSEL. R1 and R2 share an area address but have different system IDs.

IS-IS runs over Layer 2

Classic IS-IS neighbor control packets are encapsulated directly over the data-link layer (CLNS), not sent as IP packets to TCP or UDP port numbers. Integrated IS-IS additionally advertises IP reachability, allowing your PCs to communicate over IPv4.

Why metric-style wide?

Your routers use metric-style wide so IS-IS can advertise extended, larger link metrics via modern TLVs. This is appropriate for the R2 metric-change experiment (raising R2-to-R6 link cost from 10 to 40).

3. Level 1, Level 2 and area boundaries

Type	What it does	Lab example
Level 1	Maintains a topology for its own area.	R1 with R2; R5 with R4.
Level 2	Maintains inter-area backbone topology across the L2 domain.	R2, R3, R4 and R6.
Level 1-2	Participates in two separate LSDBs and connects local L1 routing to the L2 backbone.	R2, R4 and R6.

Adjacency requirements

On an Ethernet interface, neighbors must have compatible IS-IS levels and interface/network settings. For a Level 1 adjacency, they must also share an area address; Level 2 neighbors can have different area addresses. Matching the local process tag between different routers is not required.

Your verified adjacencies

Link	Adjacency type	Why
R1 - R2	L1	Same 49.0001 area
R2 - R3	L2	Different areas; backbone connection
R3 - R4	L2	Different areas; backbone connection
R4 - R5	L1	Same 49.0003 area
R3 - R6	L2	Different areas; backbone connection
R2 - R6	L2	Different areas; redundant link
R6 - R4	L2	Different areas; redundant link

ATT bit (Attached bit)

An L1/L2 router may set its ATT bit in its Level 1 LSP to indicate access to the L2 backbone. An L1 router may then install a default route toward a suitable attached L1/L2 router. The ATT bit itself does not carry a list of all distant L2 prefixes.

Important limitation: inter-area prefix reachability

Level 1 and Level 2 use separate LSDBs. Do not assume every L1 LAN is automatically visible as a specific L2 prefix in every topology. Verify the actual advertised reachability, route leaking/redistribution strategy where configured, and reverse path reachability using show ip route and ping.

4. From Hello to forwarding

Step 1 - Hello packets (IIH)

Routers send IS-IS Hello (IIH) PDUs to discover neighbors, validate relevant settings, and establish adjacencies. In your R3 output, neighbors R2, R4 and R6 were all L2 UP.

Step 2 - LSP creation

Each router originates Link State PDUs (LSPs) describing its reachable neighbors and advertised prefixes. An LSP has an identifier, sequence number, checksum, remaining lifetime and flags.

Step 3 - Flooding and LSDB synchronization

LSPs are flooded within the correct level. Complete Sequence Number PDUs (CSNPs) summarize the database, while Partial Sequence Number PDUs (PSNPs) request or acknowledge specific LSPs depending on link type and implementation.

Step 4 - SPF calculation

Each router uses Dijkstra's shortest-path-first algorithm to derive its best routes from the synchronized topology graph. The summed outgoing metric of each traversed edge determines the path cost.

Step 5 - RIB, FIB and forwarding

Eligible IS-IS routes are installed in the IP routing table (RIB) according to administrative distance and route selection. With Cisco Express Forwarding (CEF), selected routes are represented in the forwarding information base (FIB) for packet forwarding.

Useful commands

```
show isis neighbors
show isis database
show isis database detail
show isis topology
show ip route isis
show ip cef 4.4.4.4
```

Reading a database: An LSP named Khalid-R6.00-00 is R6's non-pseudonode LSP. An LSP such as Khalid-R6.03-00 represents a pseudonode originated by R6 when it is DIS on a broadcast segment.

5. DIS, pseudonodes and database fields

Why do pseudonodes appear?

On IS-IS broadcast Ethernet links, a designated intermediate system (DIS) coordinates aspects of LSP database distribution and originates a pseudonode LSP representing the LAN. Unlike OSPF, IS-IS does not have a backup DIS.

Field	Interpretation
LSPID	Originating system ID + pseudonode ID + LSP fragment number
LSP Seq Num	Tracks newer versions of an LSP
LSP Checksum	Helps detect corrupted LSP information
LSP Holdtime	Remaining lifetime before an LSP expires
ATT	Attached bit: can signal backbone connectivity to L1 routers
P / OL	Partition-related bit / overload bit
*	Locally originated LSP in the Cisco show output

Example from R6

```
IS-IS Level-1 Link State Database:
Khalid-R6.00-00  *  ...  1/0/0

IS-IS Level-2 Link State Database:
Khalid-R2.00-00       ...
Khalid-R3.00-00       ...
Khalid-R4.00-00       ...
Khalid-R6.00-00  *    ...
Khalid-R6.03-00  *    ...
```

Because R6 is in Area 49.0004 and its router-to-router adjacencies are L2, it can have its own L1 LSP without other L1 neighbor LSPs in that database. The L2 database includes LSPs originated by R2, R3, R4 and R6.

Level 1 and Level 2 LSDBs are independent

An LSP in the L2 database is not automatically copied into L1. A Level 1 router such as R1 relies on its L1 routing knowledge and access to its attached L1/L2 router for inter-area routing.

6. SPF, ECMP and your metric-40 experiment

Original state: equal-cost paths

Your R2 show isis topology initially listed R4 at metric 20 with both R3 and R6 as next hops.

Path from R2 to R4	Cost	Meaning
R2 - R3 - R4	10 + 10 = 20	Equal-cost shortest path
R2 - R6 - R4	10 + 10 = 20	Equal-cost shortest path

IS-IS therefore found two shortest paths to the same destination router. Whether both become forwarding next hops for a specific IP prefix is confirmed separately with show ip route <prefix>.

Metric optimization performed on R2

```
configure terminal
interface GigabitEthernet2/0
 isis metric 40 level-2
end
```

Destination	Best path after change	SPF metric
R4	R2 - R3 - R4	20
R6	R2 - R3 - R6	20
R6 over its direct interface	R2 - R6	40 (not best)

The direct R2-to-R6 adjacency stays UP, but it no longer wins the SPF calculation from R2. A high metric does not shutdown a link; it makes that direction less attractive.

Directional metrics

The isis metric configured on R2 changes the cost that R2 advertises for its outgoing link. R6 may advertise a different cost in the reverse direction, creating asymmetric forwarding paths.

Do not confuse these three numbers

Configured interface metric: 40 on R2 Gi2/0. SPF metric to R6: 20 via R3. Route metric to an IP prefix: may include the destination prefix advertisement metric (for example, R4 Loopback0 was observed with route metric 30).

7. Redundancy, traceroute and failover

What the observed traceroute demonstrated

When R6 was unavailable during testing, PC1 reached PC3 through the remaining chain:

```
PC1 -> R1 -> R2 -> R3 -> R4 -> R5 -> PC3

1  192.168.1.1  (R1)
2  10.0.10.2    (R2)
3  10.0.20.2    (R3)
4  10.0.30.2    (R4)
5  10.0.40.2    (R5)
6  192.168.2.10 (PC3)
```

This verifies an alternate forwarding path and reachability. A successful traceroute after a failure does not by itself measure convergence time; that would require timing packet loss or control-plane events.

Suggested controlled test sequence

Phase	Test
Baseline	Verify R2-R3 and R2-R6 L2 neighbors are UP; save topology and traceroute.
Prefer R3	Apply metric 40 on R2 Gi2/0. Check R2 topology and IPv4 route to the destination.
Fail R3	Stop the R3 GNS3 node; observe adjacency removal, route updates and forwarding via R6.
Restore	Start R3; verify the neighbor returns and SPF again prefers the lower-cost path.

On-router verification

```
show isis neighbors
show isis topology
show ip route 4.4.4.4
show ip route 192.168.2.0
show clns interface GigabitEthernet2/0
```

Traceroute final response: VPCS uses UDP probes. At the destination, ICMP Type 3 / Code 3 (destination port unreachable) commonly indicates successful termination of the trace; it does not mean IP reachability failed.

8. Architecture checklist and references

Topology checklist

Check	Expected result
NET uniqueness	Different 6-byte system IDs on all six routers
Area grouping	R1/R2 share 49.0001; R4/R5 share 49.0003
R3 L2-only	L2 neighbors R2, R4 and R6
R2 area boundary	L1 adjacency to R1; L2 adjacencies to R3 and R6
R4 area boundary	L1 adjacency to R5; L2 adjacencies to R3 and R6
Wide metrics	metric-style wide in each router IS-IS process
Path steering	R2 Gi2/0 metric 40; R3 route preferred where expected
Reachability	PC1, PC2 and PC3 can ping across configured LANs

Common troubleshooting questions

Neighbor missing? Check link state, interface IP and IS-IS activation, level compatibility, area compatibility for L1, and timers/authentication if configured.

Topology shows 20 instead of 40? 40 is the outgoing interface metric; the topology shows the shortest path. R2 may reach R6 through R3 at cost 20.

Loopback route shows metric 30? The cost to the advertising router plus its advertised prefix metric can produce 30.

Can two areas communicate? Ensure L1 routers have an appropriate route toward attached L1/L2 routers and that forward and reverse reachability to the LAN prefixes exist.

Learning outcomes

After this lab, you should be able to explain: IIH adjacency discovery, LSP origination/flooding, L1 versus L2 LSDBs, NET structure, DIS/pseudonodes, Dijkstra SPF, ECMP, directional link metrics and rerouting after an outage.

Technical references for further reading

ISO/IEC 10589 (IS-IS); RFC 1195 (integrated IS-IS); RFC 5305 (IS-IS traffic engineering extensions and wide metrics). These are reference standards, not a claim that the lab implements every optional extension.

Document note: The topology, addressing, metric change and verification examples are based on the lab details and CLI outputs shared for this project. This is an educational architecture guide, not an independently audited production network design.